

3.5.1 Eclipse Perspective

You can switch to the LogCat view in Eclipse by selecting the DDMS (Dalvik Debug Monitor Server) button on the top right corner of your Eclipse environment. Open it by selecting DDMS from Window > Open Perspective in the Eclipse menu. DDMS is the connection between the running application on the device and Eclipse. You can define filters for LogCat as well. Click on the little Green plus icon, which will open the filter dialog. This will open a separate window within the LogCat which will show results based on your filter parameters.

3.5.2 Command Line

Type the following command in your terminal window to view LogCat:

```
[user:~]> adb logcat
```

This gives the tail of the current LogCat and will keep updating as the device generates entries. For a list of command line syntaxes, visit: <http://developer.android.com/guide/developing/tools/adb.html#logcatoptions>.

3.6 Threading in Android

A Thread is a sequence of instructions executed concurrently. A CPU can process only one instruction at a time, but operating systems can interleave them on a single CPU. Based on thread priorities, the operating system allots time share to the threads. Built on the Linux platform, Android is in principle capable of running multiple threads simultaneously. However, a developer needs to know how applications use threads, in order to develop efficient applications.

3.6.1 Single Thread

Android applications run on a single thread by default, i.e. they run all commands serially. A successor is executed only upon completion of the current thread. Hence each call is blocking. The UI thread is responsible for drawing all elements on the screen, and also for processing all user inputs such as touch, pinch, clicks etc.

Running `StatusActivity` on a single thread creates problems during network calls to update status. The time for execution is not in our control, and the application cannot respond until the network call is completed, and Android may push to kill such a non-responsive thread; even if it's due to internet latency. This opens the Application Not Responding (ANR) dialog.